

# ★ Vectors

Data in tabular form  $\rightarrow$  rows & columns

## ★ Operations

### ① Addition (Dot Product)



### ② Scalar Multiplication



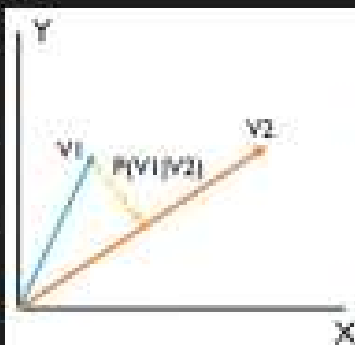
Positive  $k$   
Grows



Negative  $k$   
shrinks

### ③ Projection

Shadow a vector places on another vector.



$\Rightarrow$  We get to know unknown gestures  $v_2$  from more than  $v_1$

# \* Matrices

## ① Matrix form from equations

Suppose we have 2 equations

$$2x + 3y = 10$$

$$4x + y = 18$$

This in Matrix Form is:

$$\begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 18 \end{bmatrix}$$

## ② Addition of Matrices

## ③ Subtraction of Matrices

## ④ Multiplication of Matrices

## ⑤ Transpose $\rightarrow$ to change dimensionality

## ⑥ Determinant $\rightarrow$ Product of Eigen Values

scalar value of Matrix

## ⑦ Inverse of Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \rightarrow 2 \times 2$$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} \begin{vmatrix} e & f \\ h & i \end{vmatrix} & \begin{vmatrix} d & f \\ g & i \end{vmatrix} & \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ \begin{vmatrix} b & c \\ h & i \end{vmatrix} & \begin{vmatrix} a & c \\ g & i \end{vmatrix} & \begin{vmatrix} a & b \\ g & h \end{vmatrix} \\ \begin{vmatrix} a & c \\ d & f \end{vmatrix} & \begin{vmatrix} a & e \\ d & h \end{vmatrix} & \begin{vmatrix} a & b \\ d & e \end{vmatrix} \end{bmatrix} \rightarrow 3 \times 3$$

## ⑧ Vectors as Matrix

$$v1 = 3x + 4y$$

$$\begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} x & y \end{bmatrix} = v$$

## ⑨ VAM operations

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \quad \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} 1 & 0 \\ m & 1 \end{bmatrix}$$

Scaling Shearing

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Rotation

## ⑩ Row Echlon Method (Row manipulation)

# ★ Eigen Vectors

↳ do not change directions even if transformation applied to them.

# ★ Multivariate Calculus

## ① Differentiation

- Power Rule
- Sum Rule
- Product Rule
- Partial Differentiation

|                                                            |                                                        |                                                                                                     |
|------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| An Example                                                 |                                                        | $f(x, y, z) = x^2 + 3y + 4xz^2$                                                                     |
|                                                            | <del>Will remain the same</del>                        | Since there is no $x$ in the $3y$ term, it becomes a constant which is 0 for differentiation.       |
|                                                            | $f(x, y, z) =$<br><del>Will remain the same</del><br>1 | For $x$ and $z$ terms do not have any $y$ terms, and become a constant, so 0 after differentiation. |
|                                                            | <del>Will remain the same</del><br>8xz                 | The $x$ and $y$ terms do not have $z$ , so they become constant, and 0 after differentiation.       |
| Total differentiation: $f'(x, y, z) = 2x + 4z^2 + 8xz + 0$ |                                                        |                                                                                                     |

## ② Jacobian

↳ helps find Global Maxima of datasets

## ③ Gradient Descent (notes)